

Contents

1	Functions and Limits	2
1.1	Math 140 Review	2
1.2	The Limit of a Function	3
1.3	Calculating Limits using Limit Laws	5
1.4	Precise Definition of a Limit	6
1.5	Continuity	6
2	Derivatives	6
2.1	Derivatives and Rate of Change	7
2.2	Derivative as a Function	7
2.3	Differentiation Formulas	7
2.4	Derivatives of Trigonometric Functions	7
2.5	The Chain Rule	7
2.6	Implicit Differentiation	7
2.7	Rates of Change in Sciences	7
2.8	Related Rates	7
2.9	Linear Approximation and Differentials	7
3	Applications of Derivatives	7
3.1	Maximum and Minimum Values	7
3.2	The Mean Value Theorem	7
3.3	How Derivatives Affect the Shape of a Graph	7
3.4	Limits at Infinity	7
3.5	Summary of Curve Sketching	7
3.6	Graphing with Calculus	7
3.7	Optimization Problems	7
3.8	Newtons Method	7
3.9	Antiderivatives	7
4	Integrals	7
4.1	Areas and Distances	7
4.2	The Definite Integral	7
4.3	The Fundamental Theorem of Calculus	7
4.4	Indefinite Integrals and Net Change Theorem	7
4.5	The Substitution Rule	7
5	Applications of Integration	7
5.1	Areas Between Curves	7
5.2	Volumes	7
5.3	Volumes by Cylindrical Shells	7
5.4	Work	7
5.5	Average Value of a Function	7

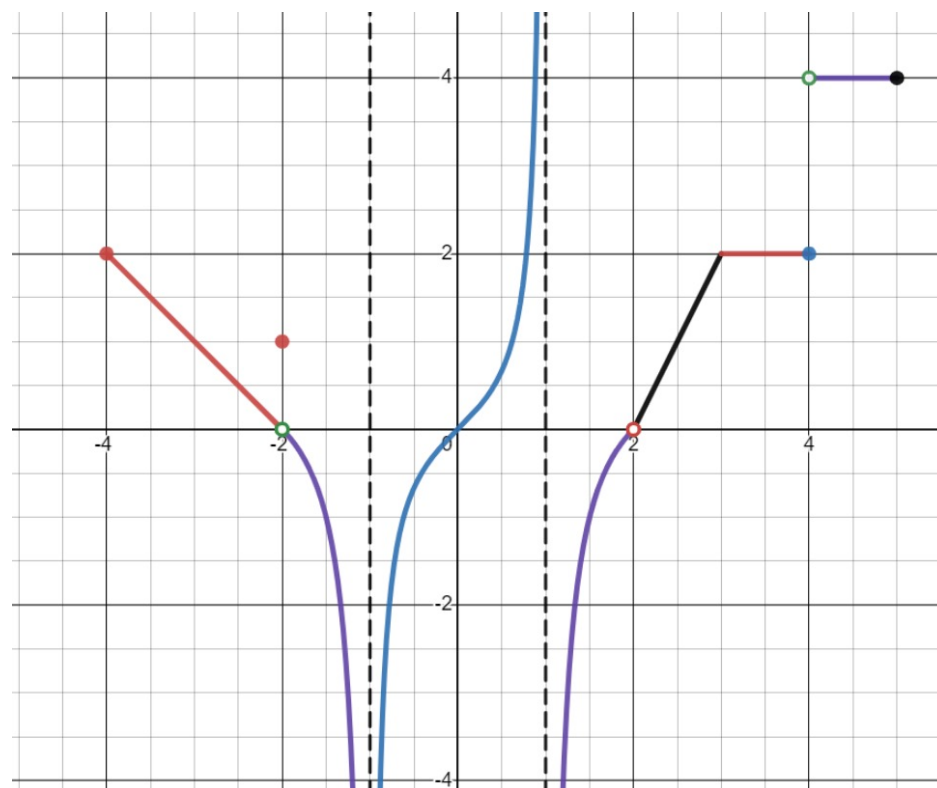
1 Functions and Limits

1.1 Math 140 Review

1. If $f(x) = \sqrt{x+5} - \sin\left(\frac{\pi}{2}x\right)$, then find the exact value of $f\left(\frac{2}{3}\right)$
2. Find the domain of each function:
 - (a) $f(x) = \sqrt{2-5x}$
 - (b) $f(x) = \frac{1}{x\sqrt{x+3}}$
 - (c) $f(x) = \sqrt[3]{2-7x}$
3. Find the difference quotient of each function:
 - (a) $f(x) = 2x^2 + x$
 - (b) $f(x) = \frac{1}{x+1}$
4. Decompose the following functions $F(x)$ into three simpler functions f, g, h so that $F(x) = (f \circ g \circ h)(x)$. Do not use the identity/trivial function for f, g or h .
 - (a) $F(x) = \sqrt{x^2 + 4}$
 - (b) $F(x) = \sin^2(2x + 1)$
 - (c) $F(x) = \tan(\cos(x^3))$

1.2 The Limit of a Function

1. The graph of $f(x)$ is drawn below. Find the following values and limits if they exist. If the limit is $+\infty$ or $-\infty$ specify which instead of “does not exist.”



(a) $\lim_{x \rightarrow -2^-} f(x)$

(i) $\lim_{x \rightarrow 1^+} f(x)$

(b) $\lim_{x \rightarrow -2^+} f(x)$

(j) $\lim_{x \rightarrow 2^-} f(x)$

(c) $\lim_{x \rightarrow -2} f(x)$

(k) $\lim_{x \rightarrow 2^+} f(x)$

(d) $f(-2)$

(l) $\lim_{x \rightarrow 2} f(x)$

(e) $\lim_{x \rightarrow -1^-} f(x)$

(m) $f(2)$

(f) $\lim_{x \rightarrow -1^+} f(x)$

(n) $\lim_{x \rightarrow 4^-} f(x)$

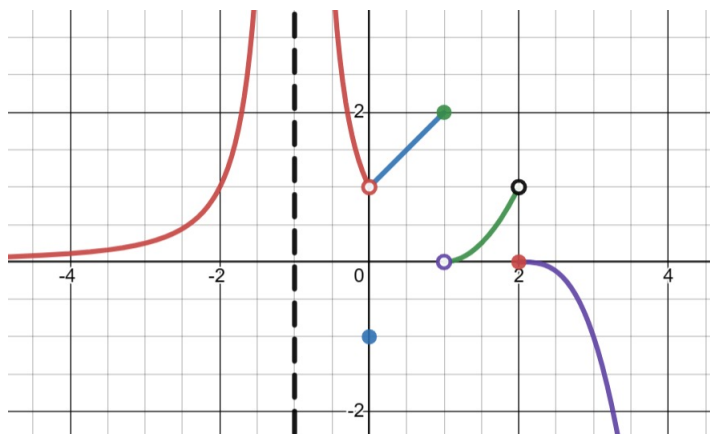
(g) $\lim_{x \rightarrow -1} f(x)$

(o) $\lim_{x \rightarrow 4^+} f(x)$

(h) $\lim_{x \rightarrow 1^-} f(x)$

(p) $f(4)$

2. Find the indicated values for the graph of $f(x)$ below. Note that there is a vertical asymptote $x = 1$. If the limit is $-\infty$ or ∞ , state which. Otherwise, if the limit doesn't exist, write DNE.



(a) $\lim_{x \rightarrow -1^-} f(x)$

(e) $\lim_{x \rightarrow 0} f(x)$

(b) $\lim_{x \rightarrow -1^+} f(x)$

(f) $\lim_{x \rightarrow 1^-} f(x)$

(c) $\lim_{x \rightarrow -1} f(x)$

(g) $\lim_{x \rightarrow 1^+} f(x)$

(d) $f(-1)$

(h) $\lim_{x \rightarrow 2} f(x)$

- (i) Give the x -values where the function $f(x)$ is discontinuous.

- (j) For the removable discontinuity/discontinuities, how can you redefine $f(x)$ to make it continuous there?

1.3 Calculating Limits using Limit Laws

1. Using the rules for limits, compute the following limits. If the limit is $\pm\infty$ state which. If the limit does not exist, write DNE. *Don't forget to keep the limit notation until you take the limit!*

(a) $\lim_{x \rightarrow 3} \frac{x^2 - 4x + 3}{x^2 - 9}$

(b) $\lim_{x \rightarrow 2} \frac{x^2 + 3x - 2}{x - 1}$.

(c) $\lim_{x \rightarrow 2^+} \frac{x^2 - 2x - 8}{x^2 - 4x + 4}$

(d) $\lim_{x \rightarrow 0} \frac{x^2 - x}{x(x + 2)}$

2. Evaluate the following limit

(a) $\lim_{x \rightarrow 3} \frac{x - 3}{\sqrt{x + 1} - 2}$.

(b) $\lim_{x \rightarrow 0} \frac{\sqrt{x^2 + 9} - 3}{x^2}$

(c) $\lim_{x \rightarrow 3^-} \frac{|x - 3| + 2}{x - 3}$ *Hint: you need to get rid of absolute values first.*

1.4 Precise Definition of a Limit**1.5 Continuity****2 Derivatives**

What is the domain of f ? What is the domain of f' ?

What is $f'(6)$?

What is the equation of the tangent line of $f(x)$ at $a = 6$?

2.1 Derivatives and Rate of Change**2.2 Derivative as a Function****2.3 Differentiation Formulas****2.4 Derivatives of Trigonometric Functions****2.5 The Chain Rule****2.6 Implicit Differentiation****2.7 Rates of Change in Sciences****2.8 Related Rates****2.9 Linear Approximation and Differentials****3 Applications of Derivatives****3.1 Maximum and Minimum Values****3.2 The Mean Value Theorem****3.3 How Derivatives Affect the Shape of a Graph****3.4 Limits at Infinity****3.5 Summary of Curve Sketching****3.6 Graphing with Calculus****3.7 Optimization Problems****3.8 Newtons Method****3.9 Antiderivatives****4 Integrals****4.1 Areas and Distances****4.2 The Definite Integral****4.3 The Fundamental Theorem of Calculus****4.4 Indefinite Integrals and Net Change Theorem****4.5 The Substitution Rule****5 Applications of Integration****5.1 Areas Between Curves****5.2 Volumes****5.3 Volumes by Cylindrical Shells****5.4 Work****5.5 Average Value of a Function**